🔷 PHASE 0 — Foundation (DO THIS FIRST, 1–2 days)

Step 0.1: Lock your research question (VERY IMPORTANT)

You are **not** "finding inhibitors".

Your final research question should be:

*Can machine learning–guided in-silico screening identify FDA-approved drugs that **modulate** NLRP3 inflammasome activity as host-directed therapy against tuberculosis?*

This single sentence:

- Aligns with TB
- Aligns with host-directed therapy
- Justifies ML
- Avoids overclaiming

📌 Write this down and don't change it later.

Step 0.2: Finalize your study design (mental clarity)

Your study has **4 pillars**:

1. **Drug repurposing** (FDA-approved drugs)
2. **NLRP3 NACHT domain** targeting
3. **Machine learning–based prioritization**
4. **Molecular docking + interaction analysis**

Nothing more. Nothing extra.

🔷 PHASE 1 — Data Collection (Pure groundwork, no ML yet)

Step 1.1: Collect FDA-approved drug dataset

You need **~200–500 compounds** (not thousands).

Sources

- DrugBank (FDA-approved only)
- PubChem (cross-check)

For each compound, store:

- Drug name
- PubChem CID
- SMILES
- Molecular weight
- LogP
- H-bond donors
- H-bond acceptors
- TPSA
- Rotatable bonds

📁 Save this as: FDA_drugs_raw.csv

👉 This becomes your **master dataset**

Step 1.2: Prepare NLRP3 protein

- Download **NLRP3 structure (NACHT domain)**
  - PDB if available
  - Else AlphaFold

Protein preparation

- Remove water
- Add hydrogens
- Energy minimization

📁 Save as: NLRP3_NACHT_prepared.pdb

This step makes your docking *legitimate*.

🔷 PHASE 2 — Build the ML Core (THIS is what upgrades your project)

Step 2.1: Create a training dataset (small but smart)

You **do NOT** need big data.

Collect from literature:

- Known NLRP3 inhibitors/modulators (MCC950, Ephedrine, EGCG, Curcumin, Berberine, etc.)

For each:

- Same molecular descriptors as FDA drugs
- Known docking scores / binding trends (from papers)

📁 Save as: NLRP3_known_binders.csv

This is your **ML training gold**.

Step 2.2: Feature engineering (simple, no overthinking)

Your ML input features:

- Molecular weight
- LogP
- TPSA
- HBD
- HBA
- Rotatable bonds

Target variable:

- Docking score (binding affinity)

No deep learning. No neural networks. **Simple = publishable.**

Step 2.3: Train your first ML model

Use:

- **Random Forest Regressor**

Why?

- Works well on small datasets
- Handles non-linearity
- Easy to explain in papers

Train model to:

Predict docking affinity to NLRP3

Evaluate using:

- R^2
- RMSE

📌 Even **moderate accuracy is acceptable** in computational biology.

Step 2.4: ML-based prioritization (key novelty)

Now apply your trained model to:

👉 FDA_drugs_raw.csv

Predict docking scores for **all FDA drugs**.

Then:

- Rank compounds
- Select **top 10–20% only**

📁 Save as: ML_prioritized_drugs.csv

🧠 **THIS is your first major novelty**

"ML-guided pre-docking screening"

🔷 PHASE 3 — Molecular Docking (now it actually means something)

Step 3.1: Dock ONLY ML-prioritized drugs

Use:

- AutoDock Vina / PyRx
- Target: NLRP3 NACHT domain

For each compound:

- Docking score
- Binding pose
- Key interacting residues

📁 Save results as:

- Docking_scores.csv

- Images of top binding poses

Step 3.2: Interaction fingerprint analysis

For top ~10 compounds:

- Hydrogen bonds
- Hydrophobic contacts
- ATP-pocket interaction

This answers:

How the drug modulates NLRP3, not just *how strongly*

🔷 PHASE 4 — ADMET + ML Clustering (research-grade polish)

Step 4.1: ADMET prediction

Use:

- SwissADME
- pkCSM

Extract:

- Oral bioavailability
- Toxicity
- Caco-2 permeability
- CYP inhibition

Add these to your dataset.

Step 4.2: Unsupervised ML clustering

Apply:

- K-Means or Hierarchical clustering

Features:

- Docking score
- ADMET parameters
- Molecular properties

Outcome:

- Cluster 1: Strong binders but toxic ❌
- Cluster 2: Moderate binders, safe ✅
- Cluster 3: Weak binders ❌

📌 This supports **host-directed therapy logic**.

🔷 PHASE 5 — TB-Specific Interpretation (THIS is what reviewers love)

Step 5.1: Controlled modulation argument

You now **explicitly argue**:

- Complete NLRP3 inhibition → harmful
- Moderate, stable binding → ideal for TB

Use:

- Binding energy range
- Interaction location
- ADMET safety

You are no longer "finding inhibitors". You are **identifying modulators**.

Step 5.2: Final compound shortlisting

Select **3–5 FDA-approved drugs** and justify each:

For each drug:

- ML rank
- Docking score
- Binding pocket
- ADMET profile
- Relevance to inflammation control

This becomes your **Results highlight table**.

🔷 PHASE 6 — Writing the Research Paper (structure matters)

Paper structure (follow EXACTLY)

1. Abstract
2. Introduction (TB → NLRP3 → HDT → ML gap)
3. Materials & Methods
    - Dataset
    - ML model
    - Docking
    - ADMET
4. Results
    - ML performance
    - Docking outcomes
    - Clustering
5. Discussion
    - TB-specific implications
    - Controlled modulation
6. Limitations
7. Conclusion